Rigidity, stress spaces, and lower bound problems on simplicial spheres

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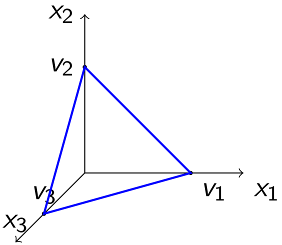
Definitions

Definition

A simplicial $(d-1)$-sphere Δ is a simplicial complex whose geometric realization $\|\Delta\|$ is homeomorphic to $\mathcal{S}^{d-1}$

Example

$\Delta = \{\{v_1, v_2\}, \{v_2, v_3\}, \{v_1, v_3\}, \{v_1\}, \{v_2\}, \{v_3\}, \emptyset\}$, with embedding $p : v_i \mapsto \vec{e_i} \in \mathbb{R}^3$.



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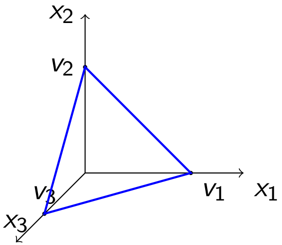
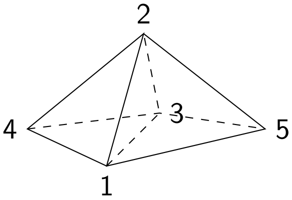


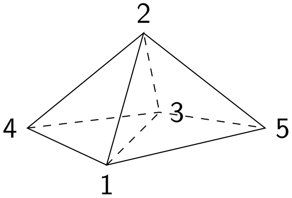
- An element $F \in \Delta$ is called an i-face if $|F| = i + 1$.
- Denote by $f_i(\Delta)$ the number of i-faces in Δ.

Two ways of describing a simplicial complex Δ



1. Describe the faces directly.
   $\Delta = \{\{1,2,4\},\{1,2,5\},\ldots,\{1,3,5\},$ and their subsets$\}$.

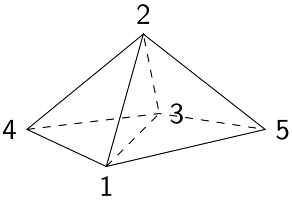
Two ways of describing a simplicial complex Δ



1. Describe the faces directly.
2. Describe the missing faces. $S \subset V(\Delta)$ is a missing face if $S \notin \Delta$ but every proper subset of S is a face of Δ.

   In this example: $\{1,2,3\}, \{4,5\}$.

Two ways of describing a simplicial complex Δ



1. Describe the faces directly.
2. Describe the missing faces.

   The simplicial complex Δ corresponds to the face ring $\mathbb{R}[\Delta] = \mathbb{R}[x_v : v \in V(\Delta)]/I(\Delta)$, where

$$I(\Delta) = (x_F = \prod_{v \in F} x_v,\ F \text{ is a missing face}).$$

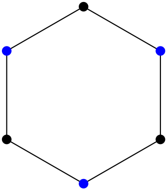
   In this example: $\mathbb{R}[\Delta] = \mathbb{R}[x_1, \ldots, x_5]/(x_4x_5, x_1x_2x_3)$.

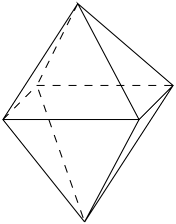
A lower bound problem

Let $S(j, d-1)$ be the class of simplicial $(d-1)$-spheres whose missing faces are at most of dimension j.

$S(d, d-1)$: all simplicial $(d-1)$-spheres.

$S(1, d-1)$: flag $(d-1)$-spheres.





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Problem: Let $\Delta \in S(j, d-1)$.
— Can we find *lower bounds* on $f_i(\Delta)$ using $f_0(\Delta), \dots, f_{i-1}(\Delta)$?
— What *tools* can we use to address this question?

The h- and g-numbers

Δ: a simplicial $(d-1)$-sphere.

- $h_i := \sum_{j=0}^{i} (-1)^{i-j} \binom{d-j}{i-j} f_{j-1}$ for $0 \le i \le d$.

  $h_i = h_{d-i}$ for $1 \le i \le d$. [Dehn-Sommerville relations]

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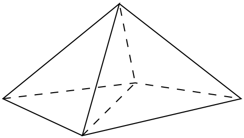
  $h_i = h_{d-i}$ for $1 \le i \le d$. [Dehn-Sommerville relations]
- $g_0 := 1$ and $g_i = h_i - h_{i-1}$ for $1 \le i \le \lceil d/2 \rceil$.
  $g_1 = f_0 - (d+1), g_2 = f_1 - df_0 + \binom{d+1}{2}$, etc.

Example 1: stacked $(d-1)$-spheres



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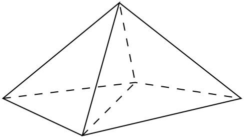

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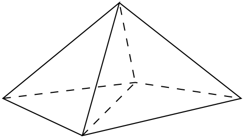

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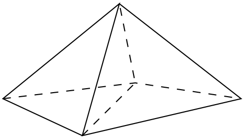

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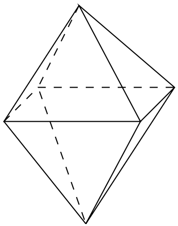
Example 2: octahedral $(d-1)$-spheres



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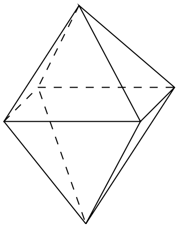

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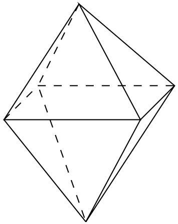

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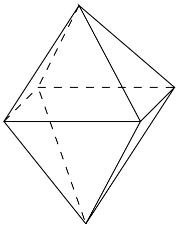

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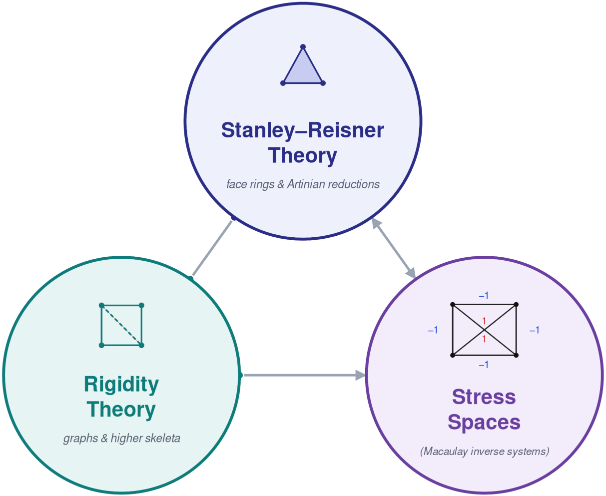
$$\gamma_2 = g_2 - (d-3)g_1 + \binom{d-2}{2} = 0.$$

A lower bound problem (rephrased)

Each f-number is nonnegative integer combinations of h-numbers and g-numbers.

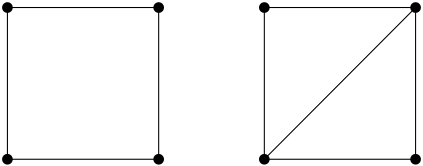
Problem: For $\Delta \in S(j, d-1)$ and $2 \leq i \leq \lfloor d/2 \rfloor$, what is the lower bound on $g_i(\Delta)$?

Tools


Stanley–Reisner Theory
face rings & Artinian reductions
Rigidity Theory
graphs & higher skeleta
−1
−1
1
1
−1
−1
Stress Spaces
(Macaulay inverse systems)

The rigidity theory

Let (G, p) be a framework in $\mathbb{R}^d$, where $G = (V, E)$ is a graph with an embedding $p : V \to \mathbb{R}^d$.
–The framework (G, p) is rigid if every sufficiently small perturbation of p that preserves edge lengths is induced by an isometry of $\mathbb{R}^d$.
–The graph G is generically rigid in $\mathbb{R}^d$ if (G, p) is rigid for a generic p.

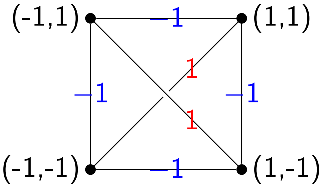


Stress spaces

Let (G, p) be a framework in $\mathbb{R}^d$.
A 2-stress on (G, p) is an assignment of weights $\{\omega_e \in \mathbb{R} : e \in E\}$ making every vertex v in balancing condition:

$$\sum_{u:uv \in E} w_{uv}(p(u) - p(v)) = \vec{0}.$$

The space of all 2-stresses on (G, p), $S_2(G, p)$, is a vector space.



The Lower Bound Theorem

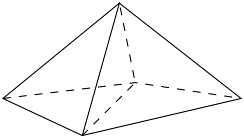
Theorem (Asimow–Roth)

If G is generically rigid in $\mathbb{R}^d$, then $\dim S_2(G) = g_2(G) \geq 0$.

Theorem (Kalai)

For $d \geq 3$, the graph of any simplicial $(d-1)$-sphere is generically rigid in $\mathbb{R}^d$.

Based on Dehn's lemma: (G, p) is rigid for the graph G of a simplicial 3-polytope with vertex embedding p.



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Theorem

Let $d \geq 4$ and let Δ be a simplicial $(d-1)$-sphere. Then

- $g_2(\Delta) \geq 0$.
- $g_2(\Delta) = 0 \Leftrightarrow \Delta$ *is stacked.*

The Hard Lefschetz Theorem

Theorem (Stanley, Adiprasito, Papadakis–Petrotou, Karu-Xiao)

Let Δ *be a simplicial* $(d-1)$*-sphere and let* $\mathbb{F}[\Delta]/(\Theta)$ *be a generic Artinian reduction of* $\mathbb{F}[\Delta]$*.*

1. $\dim(\mathbb{F}[\Delta]/\Theta)_k = h_k(\Delta)$*, for* $0 \leq k \leq d$*.*
2. *For* $\ell = \sum_{v \in V(\Delta)} x_v$*, the multiplication*

$$\cdot\ell : \mathbb{F}[\Delta]/(\Theta)_{k-1} \to \mathbb{F}[\Delta]/(\Theta)_k$$

   is injective for $k \leq \lceil d/2 \rceil$*. Hence* $\dim \mathbb{F}[\Delta]/(\Theta, \ell)_k = g_k(\Delta)$*.*

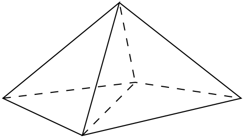
Choice of $\mathbb{F}$: $\mathbb{R}$ in the polytopal case and an infinite field of characteristic 2 in the spherical case.

The Generalized Lower Bound Theorem

Theorem

Let Δ be a simplicial $(d-1)$-sphere. Then for all $1 \leq i \leq \lfloor d/2 \rfloor$,

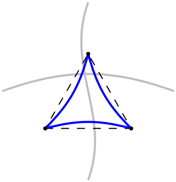
- $g_i(\Delta) \geq 0$.
- $g_i(\Delta) = 0$
  $\Leftrightarrow$ Δ *is $(i-1)$-stacked. [McMullen-Walkup, Murai-Nevo]*
  (And hence $\Delta \notin S(d-i, d-1)$.)
  $\Leftrightarrow g_{i-1}(\Delta) = \#$ *of missing $(d-i+1)$-faces.*
  [Murai-Novik-Z]



The Charney-Davis conjecture

Conjecture (Hopf–Chern–Thurston–Singer...)

Let M be a non-positively curved $2k$-manifold with Euler characteristic $\chi(M)$. Then $(-1)^k\chi(M) \geq 0$.



Geodesic triangle on a non-positively curved space (thinner).

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Conjecture (Charney-Davis)

Let Δ be a flag $(2k-1)$-sphere. Then

$$\gamma_k := (-1)^k (h_0 - h_1 + \cdots + h_{2k}) = g_k - g_{k-1} + \cdots + (-1)^k g_0 \geq 0.$$

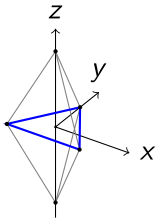
The conjecture was verified for $k = 2$ using ℓ^2-cohomology [Davis-Okun].

Current status

Let the join of two complexes Γ_1, Γ_2 be

$$\Gamma_1 * \Gamma_2 = \{\tau_1 \cup \tau_2 : \tau_1 \in \Gamma_1, \tau_2 \in \Gamma_2\}.$$

Denote by C a cycle, by $\partial\sigma^d$ the boundary of a d-simplex.



$\partial\sigma^1 * \partial\sigma^2$

	Inequality	Equality cases
$S(d, d-1)$	$g_i \geq 0$	$(i-1)$-stacked
$S(d-2, d-1)$	$g_2 - 1 \geq 0$	$C * \partial\sigma^{d-2}$ or $\partial\sigma^i * \partial\sigma^{d-i}$
$S(1, 3)$	$g_2 - g_1 + 1 \geq 0$	?
$S(1, 4)$	$g_2 - 2g_1 + 3 \geq 0$	?
$S(2, 4)$	$g_2 - g_1 \geq 0$?	?

Main results

Theorem (Novik–Z, 2026, based on Murai–Novik–Z, 2024)
Let $\Delta \in S(k+1, 2k)$ with n vertices. Then both the g-sequence and its "reverse",

$$(1, g_1, g_2, \ldots, g_k) \text{ and } (1, g_{k-1}, g_{k-2}, \ldots, g_1)$$

are M-sequences.

For $k = 3$, this means

$$\sqrt{2g_1} - 1/2 \leq g_2 \leq \binom{g_1 + 1}{2}.$$

Main results

Theorem (Nevo-Chudnovsky, 2022; sharpened in Novik-Z, 2026)

Let $d \geq 5$. A sphere in $S(1, d-1)$ satisfies

$$g_2 \geq (1/2 - \delta(d))g_1,$$

where $\delta(d)$ is a function of d with $\delta(d) \to 0$ as $d \to \infty$.

Note: Gal conjectured $\gamma_2 := g_2 - (d-3)g_1 + \binom{d-2}{2} \geq 0$.

Main results

Recall that Nevo conjectured that spheres in $S(2,4)$ satisfy $g_2 - g_1 \geq 0$.

Theorem (Novik-Z, 2026)

Any sphere in $S(2,4)$ satisfies $g_2 \geq \frac{2}{5}g_1 + \frac{6}{5}$.

Main results

Theorem (Novik-Z, 2026)

Let $\Delta \in S(3, 5)$ *with* $g_3(\Delta) = 1$. *Then*

- *either* $\Delta = \Gamma * \partial\sigma^3$, *where* Γ *is a 2-sphere.*
- *or* $\Delta = \partial\sigma^2 * \partial\sigma^2 * \partial\sigma^2$.

  It is even in $S(2, 5)$.

Future directions/references

- Higher rigidity?
  – Skeletal rigidity (largely unexplored)
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References:

1. The Partition Complex: an invitation to combinatorial commutative algebra by Karim Adiprasito and Geva Yashfe.
2. Rigidity and the lower bound theorem I by Gil Kalai.
3. PL-spheres, convex polytopes, and stress by Carl Lee.

Thanks!